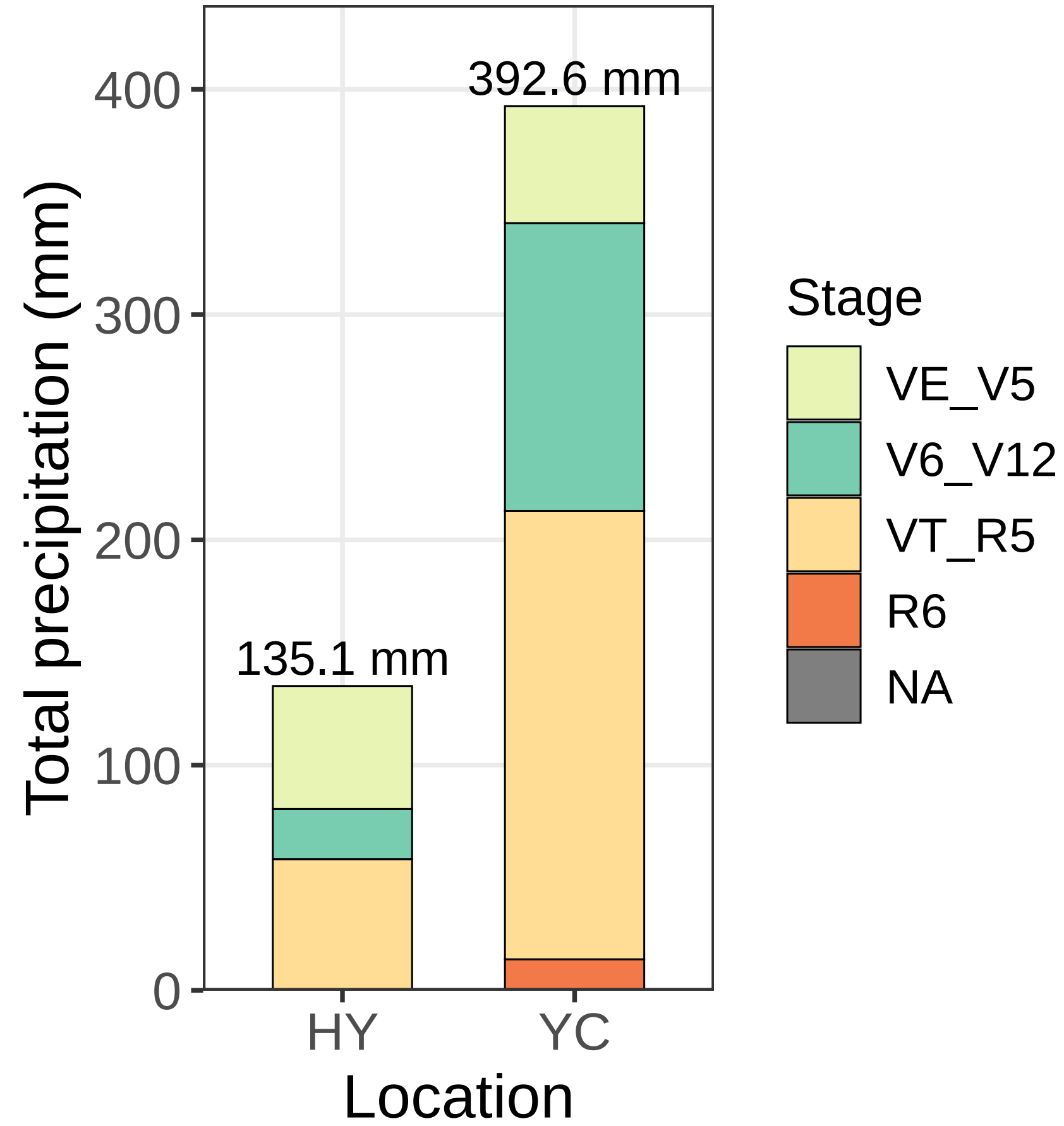al precipitation by growth stage